

Vectors

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1 Introduction

Vectors are quantities that have both *direction* and *magnitude* (how big the quantity is). Examples of vectors include *velocity* (can be positive or negative depending on the direction), *torque* (direction of torque depends on clockwise or anticlockwise spin), or *electric field* (direction depends on charge of the object). They are different from *scalars*, which only have *magnitude* but no *direction*. This includes quantities like *speed*, *temperature* or *work*.

1.1 Expressing vectors in component form

Since the arrow is on the Cartesian plane, the arrow has magnitude on the x-axis, the x-component, and it also has magnitude on the y-axis, the y-component.

This concept of components is useful for adding and subtracting vectors, which we will cover later. For now, we will just cover the method to find the

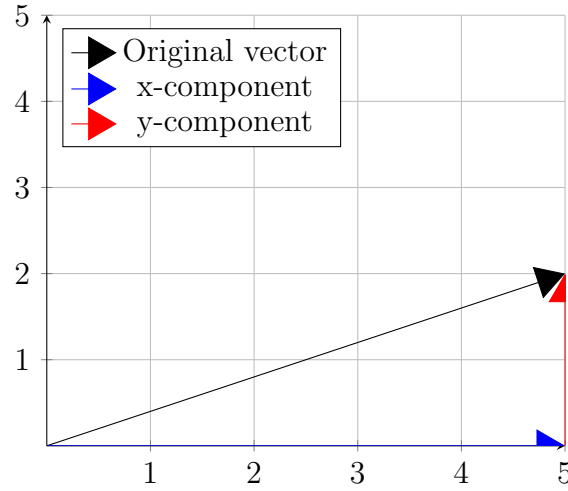


Figure 1: A vector arrow can always be split into x and y components that form a right angle triangle.

x-component and y-component. Notice that the x-component, y-component and vector arrow form a right angle triangle (1), with the x-component being the adjacent side to the angle θ and the y-component being opposite to θ .

Thus, using trigonometry,

$$\vec{v}_x = |\vec{v}| \cos(\theta) \quad (1.1)$$

and

$$\vec{v}_y = |\vec{v}| \sin(\theta). \quad (1.2)$$

We can also derive expressions of $|\vec{v}|$ and θ , where

$$|\vec{v}| = \sqrt{\vec{v}_x^2 + \vec{v}_y^2} \quad (1.3)$$

and

$$\theta = \arctan\left(\frac{\vec{v}_y}{\vec{v}_x}\right) \quad (1.4)$$

1.2 Expressing vectors in unit vector form

When the vector becomes 3-dimensional, the unit vector form can be used to quantify the vector. The expression of a vector in unit vector form uses *unit vectors*, which are vectors that have a magnitude of 1 unit, and lie exactly on the x, y, or z axis while being positive. The terms used to denote unit vectors are **i** (along the x-axis), **j** (along the y-axis), and **k** (along the z-axis).

(2)

In the unit vector form, a vector is described as the sum of unit vectors. This is comparatively easier to do than the component form in three dimensions, because of the 1-unit magnitude of a unit vector allowing for easier addition

Unit Vectors, $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$

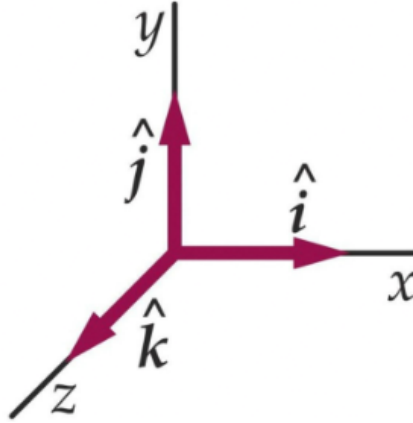


Figure 2: $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ visualised in 3 dimensions.

and subtraction of the vector, and the nature of the x, y and z components being easier to find as they are only dependent on the coefficient of 1 unit vector.

For example, if a vector $\vec{v}$ can be described by $\vec{v} = 3\mathbf{i} - 5\mathbf{j} + 2\mathbf{k}$, this means that the magnitude of the x-component is 3 (since 3×1 unit is 3 units), the magnitude of the y-component is -5, and so on. Compare that to drawing an arrow vector in three dimensional space, where finding the components using the angles between the vector and the different axes being more tedious than the unit vector method.

1.3 Expressing vectors in array form

Sometimes, vectors are represented in 1 column (also called an array), with the number of rows corresponding to the number of dimensions the vector is in (e.g. 2 rows means the vector is 2D, 3 rows means the vector is 3D, etc.). Essentially, the array form is just another way of expressing the vector in unit vector form, but in a different presentation. The vector expressed in the

unit vector section would be denoted as $\vec{v} = \begin{pmatrix} 3 \\ -5 \\ 2 \end{pmatrix}$ in array form. The top

number corresponds to the magnitude of the x-component, the middle number corresponds to the magnitude of the y-component, and the last number corresponds to the magnitude of the z-component.

Conclusion

In summary,

- Vectors are quantities with both *direction* and *magnitude*
- 2D vectors can be visually represented as arrows, with the arrowhead corresponding to direction and the length of the arrow corresponding to its magnitude.
- Furthermore, they can be split into x-components and y-components by making use of the angle between the vector and the x/y axis, θ
- Vectors can also be expressed by the sum of their x, y and z components, through the unit vector and array forms.

2 Vector addition and subtraction

Vector addition is the sum of vectors, where we are trying to find the sum of two vectors $\vec{a} + \vec{b}$, can be done in two ways:

1. Using the arrow vector form, the start point of $\vec{b}$ is placed at the end point of $\vec{a}$ (or vice versa, since vector addition is commutative), and a resultant vector is drawn from the start point of $\vec{a}$ to the end point of $\vec{b}$. (3)
2. The x-components, y-components and z-components of the vectors can be added up together to find the resultant vector.

Of these two methods, 2 will be used more as it doesn't need graph drawing, the vectors don't need to be 2D, and any number of vectors can be added easily.

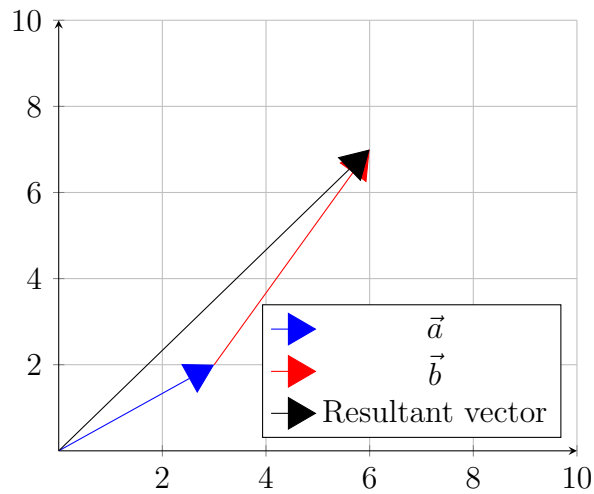
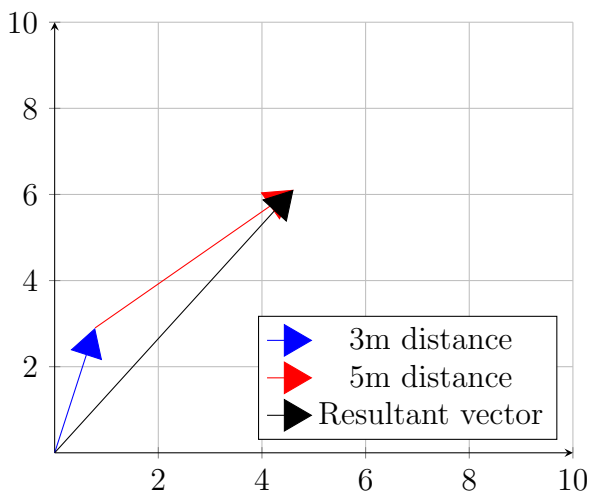


Figure 3: One can align the vector arrows to find the resultant vector.

Worked Example 1

John walks 3m 75° above East, and then 5m 40° above East. What is the total displacement of John from his original position, and what is his angle θ to the East line now?

By drawing a diagram of where John has moved, we can determine how to use the equations outlined in 1.1.



If the positive x-direction is set to East, and the positive y-direction is North, then John has a

$$+(3 \cos(75^\circ) + 5 \cos(40^\circ)) = 4.61\text{m}$$

x-displacement, and a

$$+(3 \sin(75^\circ) + 5 \sin(40^\circ)) = 6.11\text{m y-displacement. With Equation (1.3),}$$

$$s = \sqrt{4.61^2 + 6.11^2}$$

$$= 7.65\text{m}$$

With Equation (1.4),

5

$$\theta = \arctan\left(\frac{6.11}{4.61}\right) = 53.0^\circ.$$

Thus, John moved 7.65m from his original position, with a degree of 53.0° to the East line.

This can also be done in unit vector and array form, and is also much simpler to carry out as the x, y and z components are already separated. For example, if $\vec{v}_1 = 6\mathbf{i} - 2\mathbf{k} + 8\mathbf{j}$ and $\vec{v}_2 = -5\mathbf{i} - 4\mathbf{k} - \mathbf{j}$, then

$$\vec{v}_{total} = (6 - 5)\mathbf{i} + (-2 - 4)\mathbf{k} + (8 - 1)\mathbf{j}$$

$$= \mathbf{i} - 6\mathbf{k} + 7\mathbf{j}.$$

Vector subtraction operates on the same principles as vector addition, and again, has primarily two methods when solving for $\vec{a} - \vec{b}$, when $\vec{a}$ and $\vec{b}$ are both vectors:

1. When using the arrow vector form, both $\vec{a}$ and $\vec{b}$ have the same start points, and the resultant vector is drawn from the end point of $\vec{b}$ to the end point of $\vec{a}$. (4)

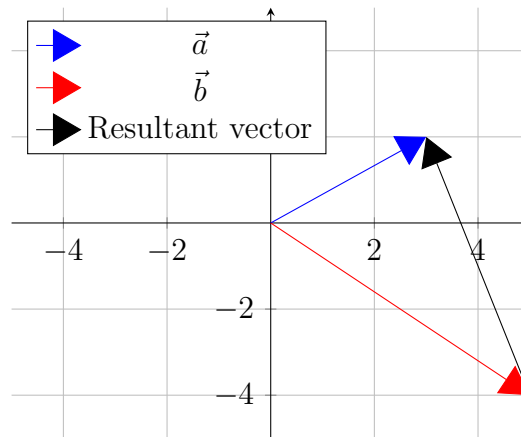


Figure 4: The method of aligning the vector arrows for addition also works for subtraction.

2. The x, y, and z-components of $\vec{b}$ can be subtracted from $\vec{a}$ to get the resultant vector.

Since the subtraction of a vector is just the addition of an equal magnitude, and opposite direction vector, one can also rewrite the expression $\vec{a} - \vec{b}$ as $\vec{a} + \vec{b}_{opp}$, which exemplifies how vector addition and subtraction are connected together. Ultimately, the concept of vector addition and subtraction involve simple ideas, but they are integral to the solving of more complex problems later in these notes.

3 Dot product

The dot product, or scalar product of two vectors is a scalar variable that is defined as

$$|v| = \vec{v}_1(\vec{v}_2) \cos \theta = |\vec{v}_1||\vec{v}_2| \cos \theta \quad (1.5),$$

where

- $|v|$ is the scalar variable,
- $\vec{v}_1$ is the first vector,
- $\vec{v}_2$ is the second vector,
- and $\cos \theta$ is the angle between the two vectors.

It can also be written as $|v| = \vec{v}_1 \cdot \vec{v}_2$, which is why the scalar product can also be referred to as the **dot** product.

To simplify this concept, the dot product is taking the "shadow" of one vector superimposed on the other (5), and multiplying the magnitude of that shadow with the magnitude of the second vector.

2. The x, y, and z- The dot product is useful as many physical quantities like

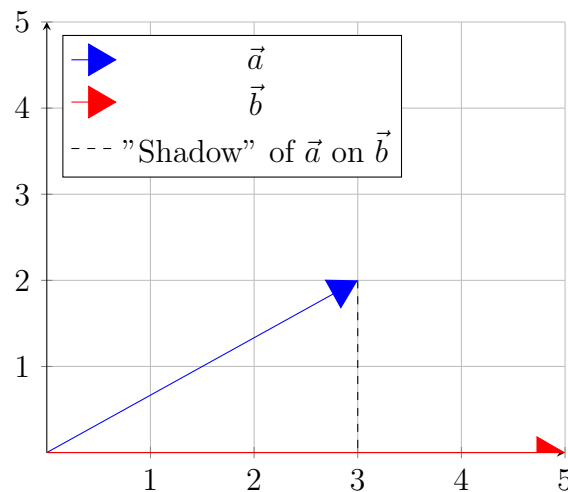


Figure 5: The dot product is the multiplication of the "shadow" of $\vec{b}$ on $\vec{a}$ with the magnitude of $\vec{a}$.

work ($W = F \cdot s$) and magnetic flux ($\phi = B \cdot A$) are dot products, and thus need intimate understanding of the dot product in order to solve problems regarding these quantities.

The dot quantity can be solved for in each of the forms mentioned in this chapter.

1. With the arrow vectors, the magnitudes of the arrows can be multiplied together, along with the cosine of the angle between the two vectors. Besides finding the scalar product, the angle θ can be found if the magnitudes of the scalar product and the two other vectors are already known

2. With unit vectors, two vectors $\vec{a}$ and $\vec{b}$ can be multiplied as if they were both polynomial expressions. For example, $\vec{a} = 6\mathbf{i} - 2\mathbf{k} + 8\mathbf{j}$ and $\vec{b} = -5\mathbf{i} - 4\mathbf{k} - \mathbf{j}$,

then

$$\begin{aligned} |v| &= (6)(-5)\mathbf{ii} + (6)(-4)\mathbf{ik} + (6)(-1)\mathbf{ij} + (-2)(-5)\mathbf{ik} + (-2)(-4)\mathbf{kk} + \\ &(-2)(-1)\mathbf{kj} + (8)(-5)\mathbf{ij} + (8)(-4)\mathbf{kj} + (8)(-1)\mathbf{jj} \\ &= -30\mathbf{ii} - 14\mathbf{ik} - 46\mathbf{ij} + 8\mathbf{kk} - 30\mathbf{kj} - 8\mathbf{jj} \end{aligned}$$

However, due to the fact that $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are all perpendicular to each other, all the terms with two different unit vectors (e.g. $\mathbf{ij}$) will be zero, as $\cos(90^\circ)=0$. Similarly, only the $\mathbf{i(i)}$, $\mathbf{j(j)}$ and $\mathbf{k(k)}$ terms will be non-zero, as the angle between one unit vector and itself is zero by definition and $\cos(0^\circ)=1$.

Thus, $|v| = -30 + 8 - 8 = -30$.

3. With the array form, a similar logic to the unit vector form can be applied, where only components on the same dimension can be multiplied

with one another. In other words, the scalar product on $\vec{v}_1 = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$

with $\vec{v}_2 = \begin{pmatrix} e \\ f \\ g \end{pmatrix}$ can be easily found by multiplying the vector components

on one vector (e.g. a), to the corresponding vector component (e.g. e), and then adding the multiplied components together to get the magnitude of the scalar product.

4 Cross product

The cross product, or vector product, is defined as

$$\vec{v} = \vec{v}_1(\vec{v}_2) \sin \theta = \vec{v}_1 \times \vec{v}_2 \quad (1.6).$$

There are a few key differences to take note of when comparing the cross product to the dot product.

1. Unlike the dot product, the cross product is a vector. Thus, it also has a direction in addition to its magnitude.
2. The direction of the cross product is perpendicular to the plane of the two vectors, and depends on which system (left-hand vs right-hand) you use when finding the cross product.
3. The cross product is also anti-commutative, where $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$. In other words, if the order of the vectors are swapped, the cross product has the opposite direction. One can observe this by using their right hand with the thumb, index and middle finger all perpendicular to each other like in 6, and observe the direction of the index and middle fingers (original vectors) when the direction of the thumb is flipped (resultant vector).

When finding the cross product with arrow vectors, there are two possible systems one can use: right-hand system (6) where $\mathbf{i} \times \mathbf{j} = \mathbf{k}$, or the left-hand system (7), where $\mathbf{i} \times -\mathbf{j} = \mathbf{k}$. Generally, the right hand system is used in physics as it aligns with the empirically derived direction of the cross product in most cases, so it is important to take note of that when solving for the cross product in physics questions.

Additionally, one can also solve a cross product using the 3×3 determinant of a vector. For example, with the vector $\vec{a} = 6\mathbf{i} - 2\mathbf{k} + 8\mathbf{j}$ and $\vec{b} = -5\mathbf{i} - 4\mathbf{k} - \mathbf{j}$, the determinant can be drawn:

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 6 & -2 & 8 \\ -5 & -4 & -1 \end{vmatrix}$$

Note that the positive/negative nature of the cross product dimensions follow the rule of $(-1)^n$, where n is a number assigned to each of the dimensions starting from x (n=0 for x, n=1 for y and n=2 for z).

We can solve each of the dimensions separately. For i, it is the cross product

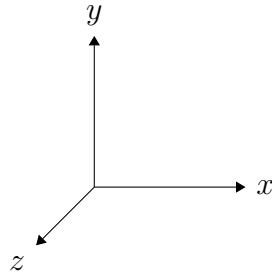


Figure 6: Right hand system.

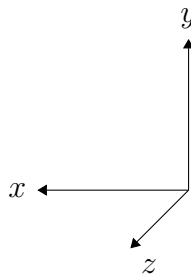


Figure 7: Left hand system.

of $\mathbf{j} \times \mathbf{k}$, so

$$\begin{aligned}\mathbf{i} &= \begin{vmatrix} -2 & 8 \\ -4 & -1 \end{vmatrix} \\ &= -2(-1) + (-4)(8) \\ &= 30\end{aligned}$$

Since $\mathbf{j}$ is equal to $-\mathbf{i} \times \mathbf{k}$,

$$\begin{aligned}\mathbf{j} &= - \begin{vmatrix} 6 & 8 \\ -5 & -1 \end{vmatrix} \\ &= -(6(-1) - (8)(-5)) \\ &= -34\end{aligned}$$

And finally, $\mathbf{k} = \mathbf{i} \times \mathbf{j}$, so

$$\begin{aligned}\mathbf{k} &= \begin{vmatrix} 6 & -2 \\ -5 & -4 \end{vmatrix} \\ &= (6(-4) - (-5)(-2)) \\ &= -34\end{aligned}$$

Thus, through using the determinant form, we have ascertained that the cross product of $6\mathbf{i} - 2\mathbf{k} + 8\mathbf{j}$ and $-5\mathbf{i} - 4\mathbf{k} - \mathbf{j}$ is $30\mathbf{i} - 34\mathbf{j} - 34\mathbf{k}$.

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